

Skill Week – 3

Task : 1

```
#include <stdio.h>

#include <stdlib.h>

#include <string.h>

int main()
{
    char command[100];

    while (1)
    {
        // Display prompt
        printf("OSSP> ");    fflush(stdout);

        // Read user input    if (fgets(command,
        sizeof(command), stdin) == NULL)
        {
            printf("\n");
            break;
        }

        // Remove newline character
        command[strcspn(command, "\n")] = '\0';

        // Handle empty input    if
        (strlen(command) == 0)
        {
            continue;
        }

        // Handle exit condition    if
```

```

(strcmp(command, "exit") == 0)
{
    printf("Exiting shell...\n");
break;
}

    // Display entered command
printf("You entered: %s\n", command);
}

return 0;
}

```

Output:

```

avulasanjana@SanjanaReddy:~$ nano main_loop.c
avulasanjana@SanjanaReddy:~$ gcc main_loop.c -o main_loop
avulasanjana@SanjanaReddy:~$ ./main_loop
OSSP> hello
You entered: hello
OSSP> ls
You entered: ls
OSSP> pwd
You entered: pwd
OSSP> test
You entered: test
OSSP> exit
Exiting shell...

```

Task: 2

```

#include <stdio.h>

#include <fcntl.h>

#include <unistd.h>

#include <stdlib.h>

int main()

```

```

{ int fd1, fd2; char
buffer[100]; ssize_t
n;

// Open File1.txt for reading fd1
= open("File1.txt", O_RDONLY);

if (fd1 < 0)
{
    perror("Error opening File1.txt");    return
1;
}

// Open File2.txt for writing
// Create it if it does not exist
// Truncate it if it already exists fd2 = open("File2.txt",
O_CREAT | O_WRONLY | O_TRUNC, 0644);

if (fd2 < 0)
{
    perror("Error opening File2.txt");
close(fd1);    return 1;
}

// Read from File1.txt and write to File2.txt
while ((n = read(fd1, buffer, sizeof(buffer))) > 0)
{
    if (write(fd2, buffer, n) != n)
    {
        perror("Error writing to File2.txt");
        close(fd1);    close(fd2);
return
1;

```

```

    }
}

if (n < 0)
{
    perror("Error reading File1.txt");
}
else
{
    printf("File copied successfully!\n");
}

// Close both files    close(fd1);
close(fd2);

return 0;
}

```

Output:

```

avulasanjana@SanjanaReddy:~$ nano File1.txt
avulasanjana@SanjanaReddy:~$ nano file_copy.c
avulasanjana@SanjanaReddy:~$ gcc file_copy.c -o file_copy
avulasanjana@SanjanaReddy:~$ ./file_copy
File copied successfully!
avulasanjana@SanjanaReddy:~$ ls
File1.txt  OSSP      docs      fork_exec.c  obj        scripts
File2.txt  README.md file_copy  include      process    src
LICENSE   assets    file_copy.c  main_loop    process.c  tests
Makefile   bin       fork_exec  main_loop.c  process.c.save

avulasanjana@SanjanaReddy:~$ cat Filed.txt
cat: Filed.txt: No such file or directory
avulasanjana@SanjanaReddy:~$ cat File1.txt
Hello, this is File1.
This content is copied using Linux system calls.
Open, Read, Write and Close are used.
avulasanjana@SanjanaReddy:~$ cat File2.txt
Hello, this is File1.
This content is copied using Linux system calls.
Open, Read, Write and Close are used.

```

Task : 3

```
#include <stdio.h>
```

```
#include <string.h>
```

```

#define MAX 100

int main()
{
    char buffer[MAX];
    int index = 0;    char ch;

    printf("Simple Keyboard Input Program\n");
    printf("Type a command (type 'exit' to quit)\n");

    while (1)
    {
        printf("OSSP> ");    index
= 0;

        while (1)
        {
            ch = getchar();

            // Enter key    if
(ch == '\n')
            {
                buffer[index] = '\0';
                break;
            }

            // Backspace key    else if
(ch == 127 || ch == '\b')
            {
                if
(index > 0)
                {
                    index--;
                }
            }
            printf("\b \b");

```

```

    }
}

// Store normal characters
else
{
    if (index < MAX - 1)
    {
        buffer[index++] = ch;
    }
}

if (strcmp(buffer, "exit") == 0)
{
    printf("Exiting Program...\n");    break;
}

printf("Command Entered: %s\n", buffer);
}

return 0;
}

```

Output:

```

avulasanjana@SanjanaReddy:~$ nano keyboard_input.c
avulasanjana@SanjanaReddy:~$ gcc keyboard_input.c -o keyboard_input
avulasanjana@SanjanaReddy:~$ ./keyboard_input
OSSP> hello
Command: hello
OSSP> ls
Command: ls
OSSP> pwd
Command: pwd
OSSP> test
Command: test
OSSP> exit
Exiting shell...

```